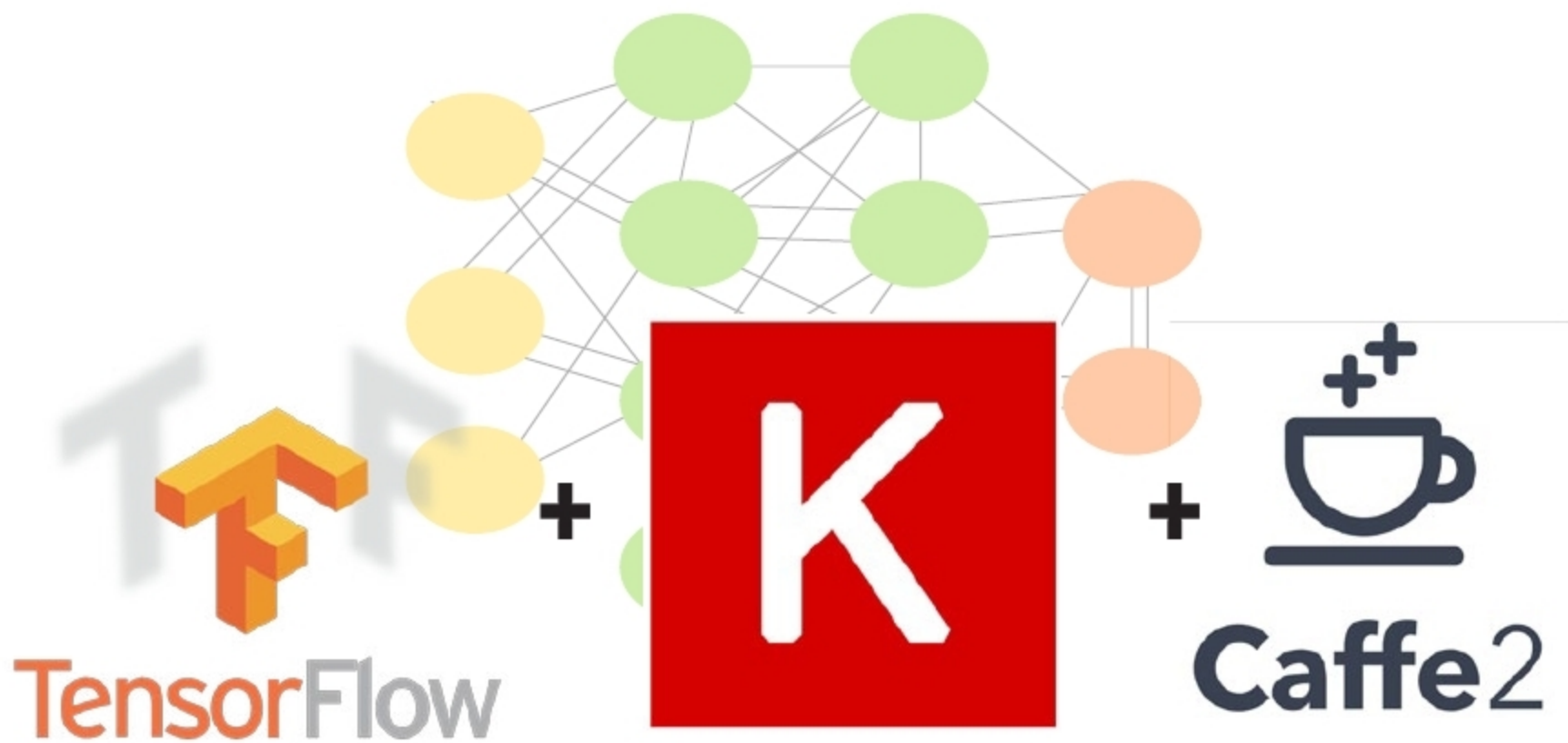


A Peep into Deep Learning Architectures for Industrial Applications

Deep learning is an approach that makes a machine imitate the network of neurons in a human brain. It consists of algorithms, which allow machines to train to perform tasks that include computer vision, speech recognition and natural language processing.



The demand for deep learning applications, such as image processing, signal recognition and data mining, in industries such as aerospace and defence, automotive, healthcare, IT and telecommunications, is expected to drive the growth of the market across the globe.

Computer vision is the most popular deep learning application used across the industry. Pattern recognition, optical character recognition, code recognition, facial recognition, object recognition, natural language processing and digital image processing are all driving the demand for deep learning.

Industry adoption of deep learning

The global deep learning market is expected to reach US\$ 18.16 billion by 2023.

In the APAC region, deep learning is growing as this technology is used not only in electronic products such as smartphones, tablets and PCs but also for medical and automotive products. The global deep learning market is poised to grow by US\$ 7.2 billion during 2020-2024, at a CAGR of 45 per cent during the forecast period. Let us now look at the drivers and characteristics of

deep learning. We will also explore the open source options for various deep learning models.

About deep learning

Advances in technology and hardware are making the development and deployment of deep learning based solutions relatively simple, and these are driving growth. There are two main branches in deep learning: discriminative modelling and generative modelling.

Discriminative modelling uses deep neural networks with hierarchical and supervised learning to predict the target. Generative

